- PROBLEM STATEMENT ID : PS9
- TEAM NAME : ZERO-DIFF
- TEAM ID : HK-092
- TEAM MEMBERS : SHUBHAM KUMAR, AKSHIT SINGH, KHUSHANK YADAV, KAMLESH MISHRA

PROBLEM & SOLUTION

Students are scattered across Github, notion, courses certification and many other platforms .

There was no Centralized dashboard to track student skill development, Different website have their own criteria of tracking

Manual portfolio review is time-consuming and inconsistent, also it is hard to differentiate beginner from skilled practioners as they generate resume by exaggerating skills without any depth knowledge.

ORBITx

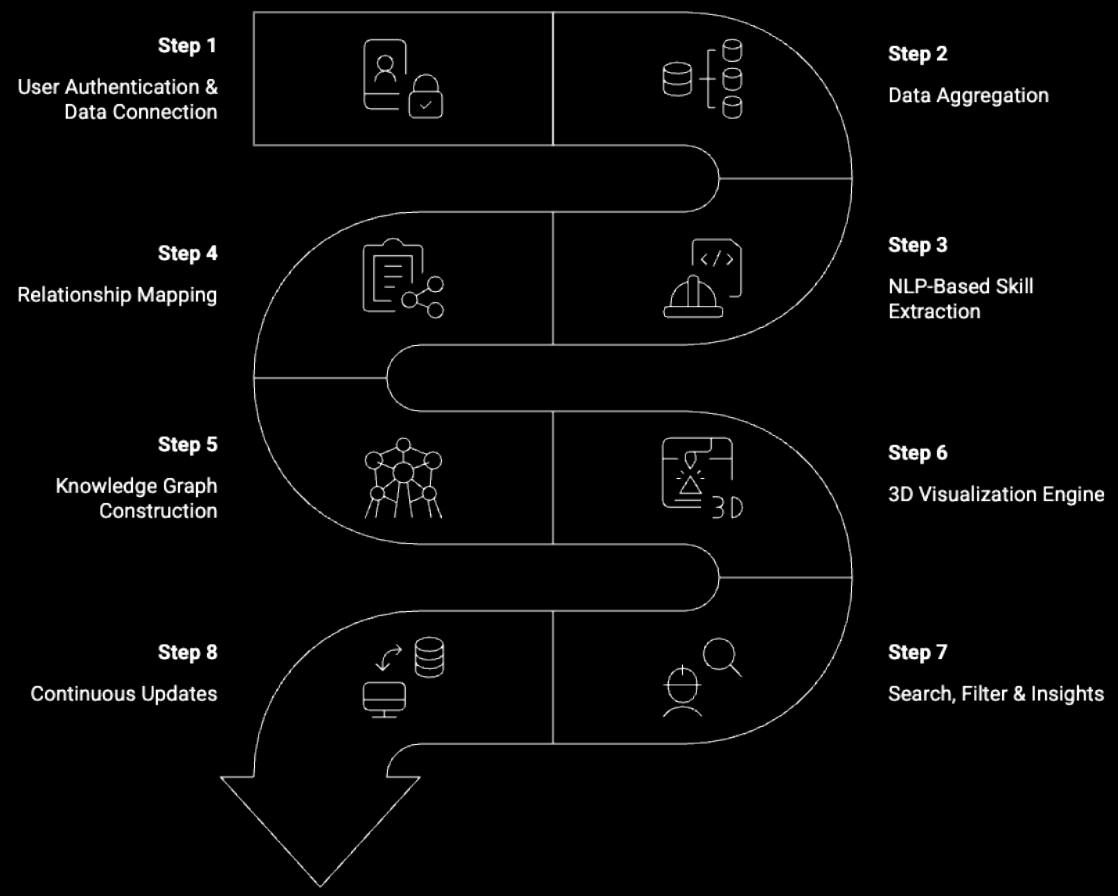
An Interactive system that automatically aggregates one's Github, Notion and Certifications into a 3D graph.

Identifies Technologies and Concepts and Builds a Relationship in the form of 3D Graph.

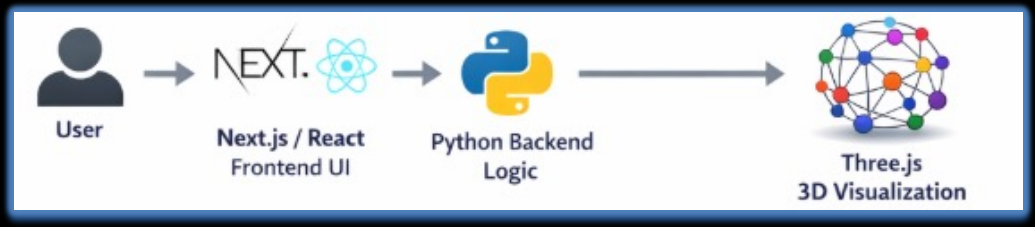
Students can explore their learning progression, skill depth, and help them to overcome the gap between who they are and who they want to become in one platform.

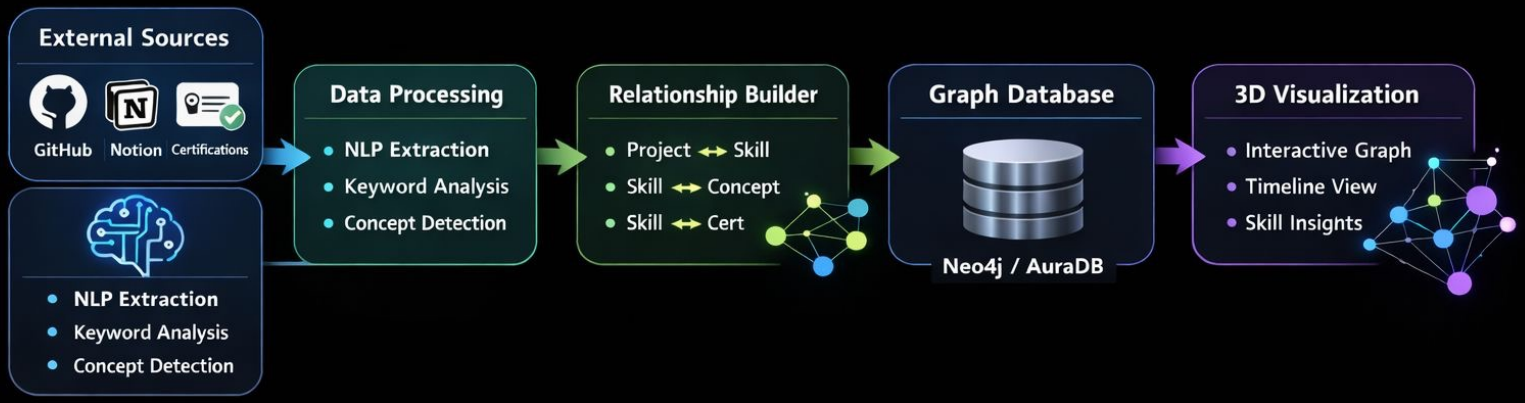
It Transformed scattered portfolios into a structured, intelligent skill ecosystem.

Flow of Solution



TECH STACK & APPROACH





UNIQUENESS & INNOVATION FACTOR

Dynamic Knowledge Graph Instead of Static Resume

Most platforms:
- Show skills as bullet points
- List projects separately
- Do not show relationships

Our system:
- Connects skills, projects, and concepts
- Shows dependencies visually
- Displays learning evolution

Interactive 3D Visualization

Most skill platforms are:
- 2D dashboards
- Static charts
- Linear timelines

We introduce:
- 3D interactive graph
- Node clustering
- Zoom & exploration

Relationship-Based Skill Intelligence

Traditional systems:
- Count skills
- Track certifications

Our system:
- Maps how skills connect
- Identifies prerequisite chains
- Detects learning progression

Skill Depth Detection

We calculate:

- Number of related projects
- Connected concepts
- Recency
- Certifications
- Dependency chains

NLP-Driven Automated Skill Extraction

Unlike manual portfolio builders:
- No manual tagging required

Skills auto-extracted from:
- GitHub repos
- README files
- Notion
- Certificates

Future-Ready Architecture

Our system can evolve into:

- AI career advisor
- Skill gap predictor
- Industry alignment scoring
- Talent matchmaking engine

Feasibility

API Availability & Standardized Data Access

GitHub and Notion provide well-documented, stable APIs with structured data formats (JSON), making data extraction reliable and implementable within limited development time.

Open-Source Ecosystem Support

Open-source NLP frameworks
Open-source graph databases
WebGL-based visualization libraries
This reduces development cost, speeds up prototyping, and ensures long-term maintainability.

Incremental Development Possibility

Phase 1: GitHub integration + basic graph
Phase 2: NLP enhancement + visualization improvements
Phase 3: AI-based insights & analytics

Challenges

Graph Scalability & Performance Optimization

As user data increases, the number of nodes and relationships grows rapidly.

Maintaining fast queries and smooth 3D visualization becomes technically demanding and requires graph optimization strategies.

Skill Standardization & Relationship Accuracy

The same skill can appear in multiple forms (e.g., abbreviations, synonyms).

Ensuring consistent skill normalization and meaningful relationships is critical to avoid inaccurate or misleading knowledge graphs.

Research & Reference

Neo4j – Knowledge Graph Applications

🔗 https://neo4j.com/use-cases/knowledge-graph/

IBM Watson – Natural Language Processing

🔗 https://www.ibm.com/watson/natural-language-processing

Three.js

🔗 https://threejs.org/